

23-S2-Q1

Q(a) GA $\xrightarrow{\text{tune}}$ PID

- (i) Real-coded representation.
- (ii) Binary ~

Q(b) 15 teams $A_1 \dots A_{15}$

3 vouchers $V_1 V_2 V_3$
\$ 60 70 80

- (i) Voucher : mini wastage.
- (ii) genotype
- (iii) fitness function

Solution (a)(i)

① Chromosome initialization

Choose a reasonable range for each parameter

eg. $k_p \in [0, 100]$ $k_i \in [0, 10]$ $k_d \in [0, 10]$

② Randomly generate each parameter within its allowable range for the initial population

③ computing fitness proportion

(1) plug each chromosome $[k_p k_i k_d]$ into PID controller

(2) Run the close-loop simulation
compute a performance metric

eg. Integral of Absolute Error

$$J = \int_0^T |e(t)| dt$$

(3) Convert this performance measure into a fitness value.

eg.
$$\text{fitness} = \frac{1}{1+J}$$

So, that lower J (better control) yields higher fitness

(4) selection

use roulette-wheel selection based on the fitness value

(5) Crossover

we can use whole arithmetic crossover

$$Z_i = \alpha X_i + (1 - \alpha) Y_i \quad (0 \leq \alpha \leq 1)$$

Z_i : child

X_i : parent 1

Y_i : parent 2.

⑤ Mutation

perturb each parameter slightly within its valid range.

eg. uniform mutation $x \rightarrow x' \in [LB_i, UB_i]$

nonuniform mutation $x' = x + N(0, \sigma)$

self - Adaptive mutation $\sigma \rightarrow \sigma'$

⑥ Replacement and Termination $x' = x + N(0, \sigma)$

form a new population

continue iterating until a stopping criterion is met

eg. fixed number of generations

or no improvement

Solution (a) (ii)

① chromosome Encoding

eg. we choose

10 bits for k_p in $[0, 100]$

8 bits for k_i in $[0, 10]$

8 bits for k_d in $[0, 10]$

eg. $\underbrace{1010001011}_{k_p} \quad \underbrace{00111001}_{k_i} \quad \underbrace{01110001}_{k_d}$

② Randomly generate bit strings.

③ Decoding

Convert each bit sub-string to a decimal integer, using

$$I(a_1 \dots a_L) = x + \frac{y-x}{2^L-1} \left(\sum_{j=0}^{L-1} a_{L-j} \cdot 2^j \right) \in [x, y]$$

get k_p k_i k_d

③ computing fitness proportions

run PID simulation

compute a performance measure

convert to fitness

use roulette wheel to obtain the
mating pool

⑤ Crossover

(1) select $n/2$ pairs of parent strings
for crossover

(2) randomly chosen pairs under mate.

(3) one-point crossover site are
randomly chosen

(4) get new population

⑥ Mutation

choose mutation rate. p_m

Alter each gene independently with p_m

⑦ Replacement and Termination

Solution (b)(i)

$$\textcircled{1} \text{ Sum of cost} = 12 + 27 + 3 + 7 + 15 + 23 + 9 + 17 + 4 + 5 + 30 + 13 + 22 + 18 + 6 = 211$$

$$\textcircled{2} \text{ Sum of Voucher} = \begin{matrix} 60 & 70 & 80 \\ V1 & V2 & V3 \end{matrix} = 210$$

Cost

30	V3	$80 - 30 = 50$
27	V3	$50 - 27 = 23$
23	V3	$23 - 23 = 0$
22	V2	$70 - 22 = 48$
18	V2	$48 - 18 = 30$
17	V2	$30 - 17 = 13$
15	V1	$60 - 15 = 45$
13	V2	$13 - 13 = 0$
12	V1	$45 - 12 = 33$
9	V1	$33 - 9 = 24$
7	V1	$24 - 7 = 17$
6	V1	$17 - 6 = 11$
5	V1	$11 - 5 = 6$
4	V1	$6 - 4 = 2$
3	not purchase	

Item A#	Cost (\$)	Voucher (V1, V2 or V3)
1	12	V1
2	27	V3
3	3	not purchase
4	7	V1
5	15	V1
6	23	V3
7	9	V1
8	17	V2
9	4	V1
10	5	V1
11	30	V3
12	13	V2
13	22	V2
14	18	V2
15	6	V1

$$V_1 (\text{limit } \$60) = 80$$

$$\text{left over} = 0$$

$$V_2 (\text{limit } \$70) = 70$$

$$\text{left over} = 0$$

$$V_3 (\text{limit } \$80) = 58$$

$$\text{left over} = 2$$

$$\text{Total left over} = 2$$

(ii) Genotype representation

① phenotype

each solution can be a string of length 15, where each item takes a value in $\{1, 2, 3, 0\}$ $[g_1 \ g_2 \ \dots \ g_{15}]$

where $g_i = \begin{cases} 1 & , \text{if item } i \text{ is purchased with } V_1 \\ 2 & , \text{if item } i \text{ is purchased with } V_2 \\ 3 & , \text{if item } i \text{ is purchased with } V_3 \\ 0 & , \text{if item } i \text{ is not purchased} \end{cases}$

Using the example (i)

$\langle 1 \ 3 \ 0 \ 1 \ 1 \ 3 \ 1 \ 2 \ 1 \ 1 \ 3 \ 2 \ 2 \ 2 \ 1 \rangle$

② genotype.

represent integer value by their binary code

phenotype	genotype
0	00
1	01
2	10
3	11

Example in (i) genotype representation

$\langle 01 \ 11 \ 00 \ 01 \ 01 \ 11 \ 01 \ 10 \ 01 \ 01 \ 11 \ 10 \ 10 \ 10 \ 01 \rangle$

(iii) Formulating the fitness function

① calculate the total cost

$$\text{cost}(V_1) = \sum_{i: g_i=1} \text{cost}(A_i)$$

$$\text{cost}(V_2) = \sum_{i: g_i=2} \text{cost}(A_i)$$

$$\text{cost}(V_3) = \sum_{i: g_i=3} \text{cost}(A_i)$$

② check feasibility

$$\text{cost}(V_1) \leq 60 \quad \text{cost}(V_2) \leq 70 \quad \text{cost}(V_3) \leq 80$$

if any sum exceed its voucher limit
we can give a large penalty (P) to
make the chromosome's fitness worse

③ compute Left over

$$L_1 = 60 - \text{cost}(V_1)$$

$$L_2 = 60 - \text{cost}(V_2)$$

$$L_3 = 60 - \text{cost}(V_3)$$

$$\text{Total Left over } L = L_1 + L_2 + L_3$$

④ fitness function

$$\text{fitness} = \frac{1}{1 + L + P}$$

$$P = \begin{cases} 0, & \text{all costs are within voucher limits} \\ 1000 \end{cases}$$